

Analyticity of Solutions to a Second Order Linear ODE

Abstract

We give a detailed derivation showing that solutions of a linear second-order ODE with real analytic coefficients are real analytic themselves. The proof proceeds by constructing a convergent power series using a Cauchy majorant argument and then extending the solution across the domain of analyticity of the coefficients. As mentioned in class, we treat a second-order ODE for ease of presentation. The methods outlined below should hold equally well for any m -th order linear ODE.

1 Introduction

Consider the ordinary differential equation

$$u''(t) + a_1(t)u'(t) + a_0(t)u(t) = 0, \quad (1)$$

with initial data

$$u(0) = \alpha_0, \quad u'(0) = \alpha_1. \quad (2)$$

Here a_0 and a_1 are assumed to be real-analytic functions convergent in the open interval $(-R, R)$.

2 Power Series Expansion

We seek a solution in the form of a power series around $t = 0$:

$$u(t) = \sum_{n=0}^{\infty} c_n t^n.$$

Differentiating,

$$u'(t) = \sum_{n=0}^{\infty} (n+1)c_{n+1}t^n, \quad u''(t) = \sum_{n=0}^{\infty} (n+2)(n+1)c_{n+2}t^n.$$

Write

$$a_i(t) = \sum_{n=0}^{\infty} a_{i,n}t^n, \quad i = 0, 1,$$

which converge for $|t| < R$. Substituting these into (1) and equating coefficients of t^n , we obtain the recurrence relation

$$(n+2)(n+1)c_{n+2} + \sum_{j=0}^n \left((j+1)a_{1,n-j}c_{j+1} + a_{0,n-j}c_j \right) = 0, \quad n \geq 0. \quad (3)$$

Thus,

$$c_{n+2} = -\frac{1}{(n+2)(n+1)} \sum_{j=0}^n \left((j+1)a_{1,n-j}c_{j+1} + a_{0,n-j}c_j \right).$$

The initial coefficients are given by $c_0 = \alpha_0$, $c_1 = \alpha_1$. Hence the c_n are determined uniquely.

3 Convergence

To prove convergence, fix $0 < r' < R$ and define

$$A_i(r') := \max_{|t|=r'} |a_i(t)|, \quad i = 0, 1.$$

From real analyticity of a_i , we get

$$|a_{i,n}| \leq \frac{A_i(r')}{(r')^n}.$$

Lemma 3.1. *There exist constants $K > 0$ and $r' \in (0, R)$ such that*

$$|c_n| \leq \frac{K}{(r')^n}, \quad n \geq 0.$$

Proof. Choose $K \geq \max(|\alpha_0|, |\alpha_1|)$. Assume inductively that $|c_m| \leq K/(r')^m$ for $m \leq n+1$. From (3),

$$\begin{aligned} |c_{n+2}| &\leq \frac{1}{(n+2)(n+1)} \sum_{j=0}^n \left((j+1)|a_{1,n-j}||c_{j+1}| + |a_{0,n-j}||c_j| \right) \\ &\leq \frac{K}{(r')^{n+1}(n+2)(n+1)} \sum_{j=0}^n \left((j+1)A_1(r') + A_0(r')r' \right). \end{aligned}$$

The sum simplifies to

$$\frac{(n+1)(n+2)}{2} A_1(r') + (n+1)A_0(r')r'.$$

Thus,

$$|c_{n+2}| \leq \frac{K}{(r')^{n+1}} \left(\frac{A_1(r')}{2} + A_0(r')r' \right).$$

Choosing r' sufficiently small ensures

$$q := r' \left(\frac{A_1(r')}{2} + A_0(r')r' \right) < 1,$$

which yields

$$|c_{n+2}| \leq \frac{K}{(r')^{n+2}}.$$

The claim follows by induction. □

Hence the series $\sum c_n t^n$ converges absolutely for $|t| < r'$ and defines an analytic solution near 0.

4 Analytic Continuation

Because a_0 and a_1 are analytic on $|t| < R$, the same argument applies around any point t_0 with $|t_0| < R$. The uniqueness of solutions to (1) ensures that local analytic solutions agree on overlaps, giving a unique global analytic solution on $(-R, R)$.